FIITJEE

ALL INDIA TEST SERIES

FULL TEST – XI

JEE (Advanced)-2021

PAPER –1

TEST DATE: 26-09-2021

ANSWERS, HINTS & SOLUTIONS

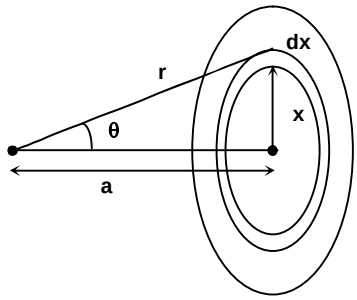
Physics

PART – I

SECTION – A

1. B

Sol. Consider a circular strip of radius x and width 'dx' of the cross section of the sphere



$d\phi = EdA\cos\theta$

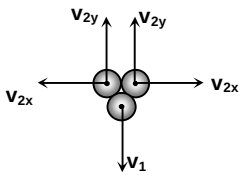
$= \frac{\rho r}{3\varepsilon_0} 2\pi x dx \frac{a}{r} = \frac{2\pi\rho a x dx}{3\varepsilon_0}$

Total flux over the cross section

$$\phi = \int d\phi = \frac{2\pi\rho a}{3\varepsilon_0}\int_0^{\sqrt{R^2-a^2}} x dx = \frac{\pi\rho a}{3\varepsilon_0}(R^2 - a^2)$$

2. C

Sol. From conservation of momentum



$mv_1 - 2mv_{2y} = 0$

$\Rightarrow v_1 = 2v_{2y}$

$h = v_1 t_1 + \frac{1}{2} g t_1^2$

$h = -v_{2y} t_2 + \frac{1}{2} g t_2^2$

$\Rightarrow h\left(\frac{1}{t_1} + \frac{2}{t_2}\right) = \frac{1}{2} g(t_1 + 2t_2)$

$\Rightarrow h = \frac{g t_1 t_2 (2t_2 + t_1)}{2(t_2 + 2t_1)}$

3. A

Sol. From conservation of mechanical energy

$$mgh = Mg\left(\sqrt{a^2+h^2}-a\right)$$

$$\Rightarrow h = \frac{2mMa}{M^2-m^2}$$

4. C

Sol. Comparing with radioactive decay

$$a = a_0e^{-\lambda t},\ \text{where}\ \lambda = \frac{\ell n2}{t_0}$$

$$\Rightarrow \frac{dv}{dt} = a_0e^{-\lambda t}$$

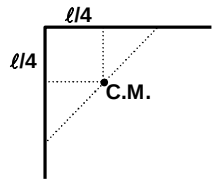
$$\Rightarrow \int_0^v dv = \int_0^t a_0e^{-\lambda t}dt$$

$$\Rightarrow v = \frac{a_0}{\lambda}(1-e^{-\lambda t})$$

Terminal velocity $v_T = \frac{a_0}{\lambda} = \frac{a_0t_0}{\ell n2}$

5. A

Sol. From conservation of angular momentum



$$mu\frac{3\ell}{4} = I_{cm}\omega$$

$$= 2\left[\frac{m\ell^2}{12}+m\left(\frac{\sqrt{2}\ell}{4}\right)^2\right]\omega$$

$$\omega = \frac{9u}{5\ell}$$

6. D

Sol. $F(t) = I\ell B = \frac{B\ell v}{R}\left(e^{-t/RC}\right)\ell B = \frac{B^2\ell^2v}{R}e^{-t/RC}$

$q(t) = \varepsilon C\left(1-e^{-t/RC}\right) = B\ell vc\left(1-e^{-t/RC}\right)$, $I(t) = \left(\frac{B\ell v}{R}\right)e^{-t/RC}$

Rate of energy stored in the capacitor P(t) = VI

$= \varepsilon\left(e^{-t/RC}\right)\frac{\varepsilon}{R}\left(1-e^{t/RC}\right) = \left(\frac{B^2\ell^2v^2}{R}\right)e^{-t/RC}\left(1-e^{-t/RC}\right)$

7. A, C, D

Sol. If $n_3 < n_1$, $\frac{n_3}{n_2} < \frac{n_1}{n_2} \Rightarrow \sin\theta'_C < \sin\theta_C \Rightarrow \theta'_C < \theta_C$

So, total internal reflection occurs at AB since $\theta > \theta_C$

If $n_3 > n_1$, the ray will refract at surface AB

If θ_1 is the angle of refraction for surface AB

$$n_3\sin\theta_1 = n_2\sin\theta \Rightarrow \sin\theta_1 = \frac{n_2}{n_3}\sin\theta$$

Since, $\sin\theta > \frac{n_1}{n_2}, \sin\theta_1 > \frac{n_2}{n_3}\frac{n_1}{n_2}$

$\Rightarrow \sin\theta_1 > \frac{n_1}{n_3} \Rightarrow \sin\theta_1 > \sin\theta_C''$

where θ_C'' is the critical angle for interface CD So, the ray will undergo total internal reflection at interface CD and will return to medium of refractive index n_2 (according to optical reversibility of light)

8. A, C

Sol. Current $I = \frac{\varepsilon}{R} = \frac{1}{R}(2\pi r)\frac{r}{2}\frac{dB}{dt} = \pi t$

Ring will topple when $\tau_B > \tau_g$

$\Rightarrow \quad I\pi r^2 B_x > mgr$

$\Rightarrow \quad \pi^2 t > mg$

$t > \frac{10}{\pi}\sec$

So, the ring will topple at $t = \frac{10}{\pi}\sec$

Heat generated $H = \int I^2 R dt = \int_0^{10/\pi} \pi^2 t^2 R dt = \pi^2 R\left[\frac{t^3}{3}\right]_0^{10/\pi} = \frac{2}{3\pi}kJ$

9. A, C

Sol. $\frac{dN}{dt} = p - \lambda N$

$\Rightarrow \int_0^N \frac{dN}{p-\lambda N} = \int_0^t dt$

$\Rightarrow N = \frac{p}{\lambda}\left(1 - e^{-\lambda t}\right)$

Number of nuclei decayed in time t

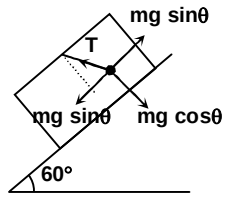
$= pt - N = p\left[t - \frac{1}{\lambda}(1 - e^{-\lambda t}\right]$

Energy released = $pE_0\left[t - \frac{1}{\lambda}\left(1 - e^{-\lambda t}\right)\right]$

Rate of energy released = $E_0\lambda N = pE_0\left(1 - e^{-\lambda t}\right)$

10. B, C

Sol. With respect to the reference frame attached to the box pseudo force mg sin θ will act on the bob up the inclined plane



From work energy theorem (In the reference frame attached to the box)

$mg\cos 60°\ell(1 - \cos 60°) = \frac{1}{2}mv^2$

$\Rightarrow v = \sqrt{\frac{g\ell}{2}}$

$T - mg\cos 60° = \frac{mv^2}{\ell} \quad \Rightarrow T = \frac{mg}{2} + \frac{m}{\ell}\frac{g\ell}{2} = mg$

11. A, B, C, D

Sol. $L = \frac{\lambda}{2} + \frac{\lambda}{2} + \frac{\lambda}{4} = \frac{5\lambda}{4} \Rightarrow \lambda = \frac{4L}{5}$

$\Delta P_0 = |\Delta P_m \cos kx| = |\Delta P_m \cos \frac{2\pi}{\lambda}\frac{L}{2}|$

$= |\Delta P_m \cos \frac{\pi L}{4L/5}| = |\Delta P_m \cos \frac{5\pi}{4}| = \frac{\Delta P_m}{\sqrt{2}}$

$A = |A_{max} \sin kx| = |A_{max} \sin\left[\left(\frac{2\pi}{\lambda}\right)\left(\frac{\ell}{3}\right)\right]| = \frac{A_{max}}{2}$

12. A, C

Sol. Let $M = kh^a G^b \eta^c$, where k is a dimensionless constant

$\Rightarrow \left[ML^0T^0\right] = \left[ML^2T^{-1}\right]^a \left[M^{-1}L^3T^{-2}\right]^b \left[ML^{-1}T^{-1}\right]^c$

$\Rightarrow a = \frac{5}{9}, b = \frac{-1}{3}$ and $c = \frac{1}{9}$

Let $L = kh^a G^b \eta^c$, where k is a dimensionless constant

$\Rightarrow \left[M^0LT^0\right] = \left[ML^2T^{-1}\right]^a \left[M^{-1}L^3T^{-2}\right]^b \left[ML^{-1}T^{-1}\right]^c$

$\Rightarrow a = \frac{1}{3}, b = 0$ and $c = \frac{-1}{3}$

SECTION – C

13. 00004.50

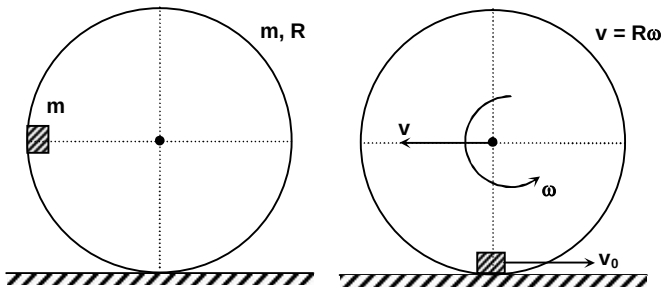
Sol. $\int f dt = mv_0 - mv$ …(i)

$\int fR dt = \frac{2}{3}mR^2\omega$ …(ii)

$v = R\omega$ …(iii)

$mgR = \left[\frac{1}{2}\left(\frac{2}{3}mv^2\right) + \frac{1}{2}mv^2\right] + \frac{1}{2}mv_0^2$

$\Rightarrow v = \sqrt{\frac{9gR}{20}}$



14. 00013.33

Range 13.32 to 13.34

Sol. Let the object is at a depth h below the water surface. When the lens is placed just above the water surface, object distance for the lens

$u = -\frac{h}{4/3} = -\frac{3h}{4}$

For the lens

$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

$\Rightarrow \frac{1}{15} + \frac{4}{3h} = \frac{1}{f}$ ...(i)

When the lens is just below the water surface, if f′ be the focal length of the lens

$$\frac{f'}{f}=\frac{(\mu-1)}{\left(\frac{\mu}{\mu_w}-1\right)}=\frac{1.5-1}{\left(\frac{1.5}{4/3}-1\right)}=4$$

$\Rightarrow f' = 4f$

Image distance for the lens

$v=\mu_w\times 60=80\,cm$

$$\frac{1}{v}-\frac{1}{u}=\frac{1}{f}$$

$$\Rightarrow \frac{1}{80}-\frac{1}{-h}=\frac{1}{4f}$$

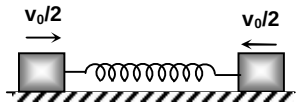
$$\Rightarrow \frac{1}{80}+\frac{1}{h}=\frac{1}{4f} \qquad ...(ii)$$

From equation (i) and (ii)

f = 13.33 cm

15. 00000.25

Sol. In centre of mass frame the block B and C will execute SHM, just after collision of block A with B. Velocities of block B and C in centre of mass frame of blocks (B + C) will be as shown.



Velocity of centre of mass of system (B + C)

$$v_{CM}=\frac{mv_0+m\times 0}{2m}=\frac{v_0}{2}$$

The block D will attain velocity v_0, if after half oscillation block C collides with block D.

$\Rightarrow x_{cm}=\ell$

$$\Rightarrow \frac{v_0}{2}\frac{T}{2}=\ell$$

$$\Rightarrow \ell=\frac{v_0}{4}2\pi\sqrt{\frac{m}{2k}}=\pi v_0\sqrt{\frac{m}{8k}}$$

$\Rightarrow n = 0.25$

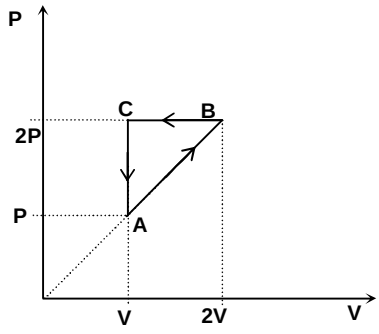
16. 00000.28

Sol. In process $A \to B$

$T \propto V^2$

$\Rightarrow PV \propto V^2$

$\Rightarrow P \propto V \Rightarrow PV^{-1}$ = constant



Molar heat capacity in this process

$$C=C_V+\frac{R}{1-\alpha}=\frac{5R}{2}+\frac{R}{1+1}=3R$$

$\Delta Q_{AB}=nC(T_B-T_A)=0.36$ kJ

$\Rightarrow 3R(T_B-T_A)=0.36$ kJ

P-V diagram of the entire process will be as shown.

$$T_C=\frac{T_B}{2} \text{ and } T_A=\frac{T_B}{4} \qquad (\text{since } T=\frac{PV}{nR})$$

$$\Delta Q_{BC}=nC_P(T_C-T_B)=\frac{7R}{2}(T_C-T_B)$$

$3R(T_B-T_A)=0.36$ kJ

$$\Rightarrow 3R\left(T_B-\frac{T_B}{4}\right)=3R\times\frac{3T_B}{4}=0.36$$

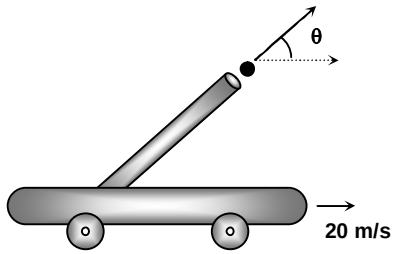
$RT_B = \frac{1.44}{9} = 0.16$ kJ

$\Rightarrow \Delta Q_{BC} = \frac{7R}{2}(T_C - T_B)$

$= \frac{7R}{2}\left(\frac{T_B}{2} - T_B\right) = -\frac{7}{4}RT_B = -\frac{7}{4} \times 0.16 = -0.28$ kJ

17. 00103.92

Sol. $\frac{u^2}{g} = 40 \Rightarrow u = 20$ m/s



If the gun is inclined at an angle θ with the horizontal, range of the shell

$R = \frac{2u_y}{g}u_x = \frac{2 \times 20\sin\theta}{10}(20\cos\theta + 20)$

$\Rightarrow R = 80\left(\frac{1}{2}\sin 2\theta + \sin\theta\right)$

For R_{max}, $\frac{dR}{d\theta} = 0 \Rightarrow \cos 2\theta + \cos\theta = 0$

$\Rightarrow \theta = 60°$

$R_{max} = 80\left(\frac{1}{2}\sin 120° + \sin 60°\right) = 80 \times \frac{3}{2} \times \frac{\sqrt{3}}{2} = 60\sqrt{3} = 103.92$ m

18. 00003.31

Sol. In the figure-(i) let distance between O of the vernier scale and zero of the main scale x_1

$\Rightarrow x_1 + 4$ M.S.D. $= 4$ V.S.D

$\Rightarrow x_1 = 4$ V.S.D. $- 4$ M.S.D. $= 4 \times \frac{11}{10}$M.S.D. $- 4$M.S.D $= 0.4$M.S.D. $= 0.4$ mm

Zero error $= -x_1 = -0.4$ mm

In the figure-(ii) let the distance between the zero of the vernier scale and 3.2 cm mark of the main scale be x_2.

$\Rightarrow x_2 + 3$ V.S.D. $= 4$ M.S.D.

$\Rightarrow x_2 = 4$M.S.D. $- 3 \times \frac{11}{10}$M.S.D. $= 0.7$M.S.D. $= 0.7$ mm

Measured thickness of the plate $= 3.2$ cm $+ x_2 = 3.2$ cm $+ 0.7$ mm $= 3.27$ cm

Actual thickness of the plate = measured thickness $-$ zero error

$= 3.27$ cm $-$ (-0.04 cm) $= 3.31$ cm

Chemistry

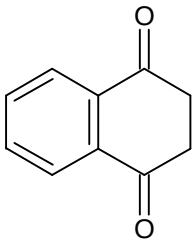
PART – II

SECTION – A

19. C

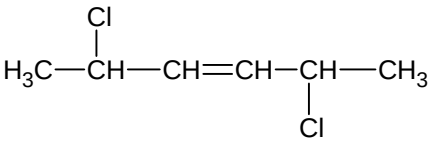
Sol. Number of geometrical isomers possible for the molecule $CH_3 - CH = CH - CH = CH - CH = CH - CH_3$ is 6.

20. D

Sol.  will tautomeris.

21. A

Sol. Number of stereo isomers for the

 is 6

$[dcd]$ $[dTd]$

$[\ell c\ell]$ $[dT\ell]$

$[dc\ell]$ $[\ell T\ell]$

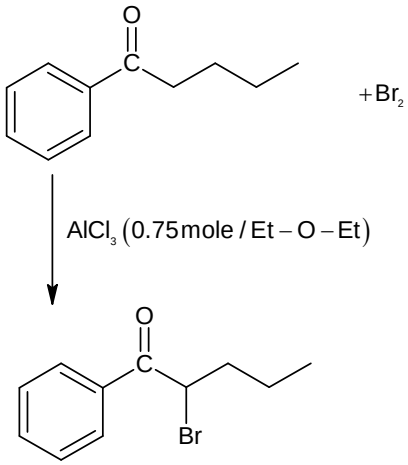
22. B

Sol. (II) is because after removing (II) hydrogen conjugate base is highly resonance stabilized compare to the other hydrogen.

23. D

Sol. pK_a values in the order of ii > iii > i > iv.

24. D

Sol. 

Enol is more reactive towards bromine than the aromatic ring.

25. B, C, D

Sol. $\left[Pt(NH_3)_4\right][PtCl_4]$, $\left[Co(en)_3\right]\left[Cr(CN)_6\right]$ and $\left[Pt(NH_3)_4\right][PtCl_6]$ shows co-ordination isomerism but $\left[Ag(NH_3)_2\right][AgCl_2]$ will not show co-ordination isomerism.

26. B, C

Sol. Complex $\left[Cu(NH_3)_4\right]^{2+}$ and $\left[Fe(CN)_6\right]^{3-}$ is paramagnetic due to presence of unpaired electron.

But due to absence unpaired electron in $\left[Ni(CO)_4\right]$ and $\left[Fe(CN)_6\right]^{4-}$ these complex are diamagnetic.

27. A, B

Sol. CdS and As_2S_3 - yellow
SnS – Brown
PbS – Black

28. B, C

Sol. $NH_4NO_2 \xrightarrow{\Delta} N_2 + H_2O$

$NH_4NO_3 \xrightarrow{\Delta} N_2O + H_2O$

$(NH_4)_2Cr_2O_7 \xrightarrow{\Delta} N_2 + Cr_2O_3 + H_2O$

29. A, C, D

Sol. $I_2O_5 + H_2O \longrightarrow HIO_3$

30. A, C, D

Sol. Aqueous solution of borax is buffer

$Na_2B_4O_7 + H_2O \longrightarrow 4H_3BO_3 + 2NaOH$

- $H_3PO_4 + NaOH \longrightarrow NaH_2PO_4 + H_2O$

  $1-1 \quad 2.5-1 \quad 1$

  $= 1.5$

- $NaH_2PO_4 + NaOH \longrightarrow Na_2HPO_4 + H_2O$

  $1-1 \quad 1.5-1 \quad 1$

  $= 0 \quad = 0.5$

Now, $Na_2HPO_4 + NaOH \longrightarrow Na_3PO_4 + H_2O$

$1-0.5 \quad 0.5-0.5 \quad 0.5$

$= 0.5 \quad = 0$

$\therefore$ Above solution is buffer solution.

- $CH_3-\overset{\overset{O}{\|}}{C}-OH + NaOH \longrightarrow CH_3-\overset{\overset{O}{\|}}{C}-ONa + H_2O$

  $2-1 \quad 1-1 \quad 1$

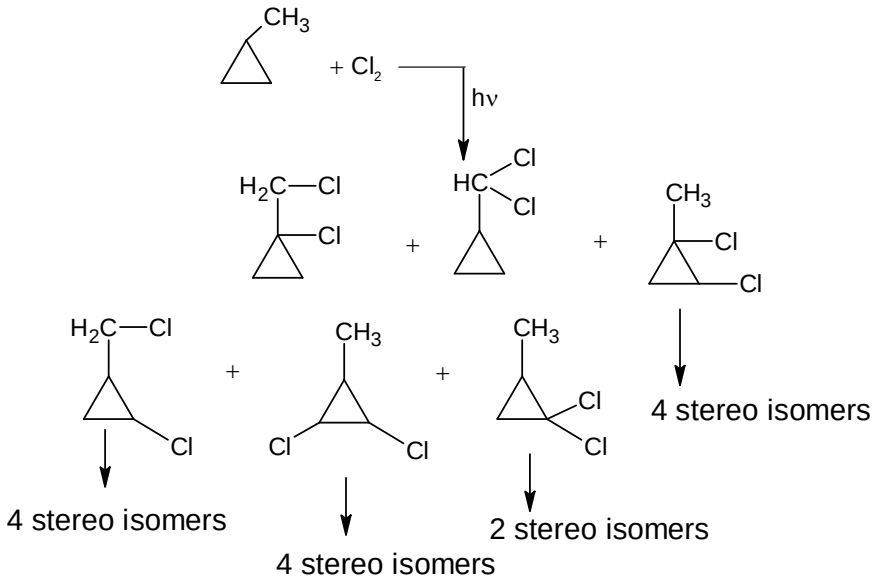
  $= 1 \quad = 0$

$\therefore$ Above solution is buffer solution.

SECTION – C

31. 00003.20

Sol.



$\therefore$ Total Dihalogen derivatives = 16

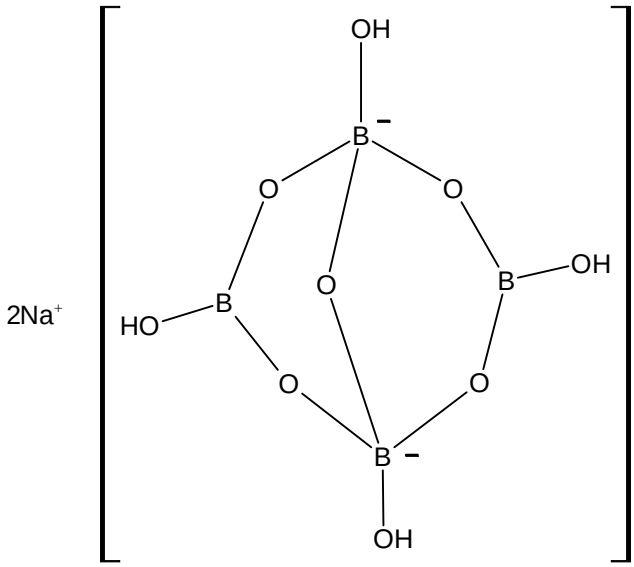
32. 00007.50

Sol. Concept of titration

$$I_2O_5 + CO \longrightarrow CO_2 + I_2$$

33. 00001.25

Sol. Borax



34. 00001.80

Sol. Real gas equation

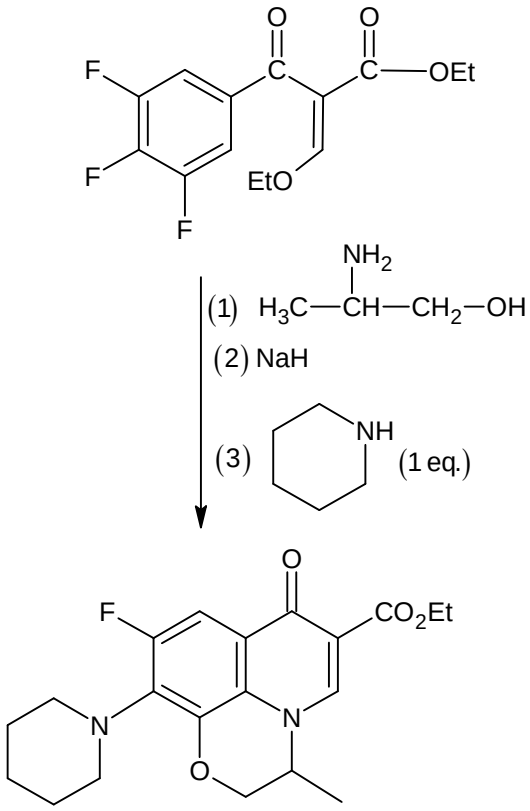
$$\left(P + \frac{an^2}{V^2}\right)(V - nb) = nRT$$

35. 00001.60

Sol. Number of stereo isomers for the complex $\left[Ma_2b_2cd\right]$ is 8.

36. 00002.50

Sol.



Above reaction is example of ArSN reaction.

Mathematics

PART – III

SECTION – A

37. A

Sol. $\lim\limits_{h\to0}\dfrac{f(2+h)}{\sin h}=3$

For finite limit, f(2) = 0

$\therefore \lim\limits_{h\to0}\dfrac{f(2+h)-f(2)}{\left(\dfrac{\sin h}{h}\right)\times h}=f'(2) \Rightarrow f'(2)=3$

38. B

Sol. $\left|\log_{\sqrt{2}}30-\left|\log_2 9+\left|\log_4 9-\left|\log_{1/2}5\right|\right|\right|\right|=\log_2 60$

39. B

Sol. $\left(\cos^2\alpha-2\cos\alpha\right)\sec^2\beta+9\,\text{cosec}^2\beta+5\sec^2\beta = \left((\cos\alpha-1)^2+4\right)\sec^2\beta+9\,\text{cosec}^2\beta$

40. C

Sol. f(f(1)) = 0 $\Rightarrow$ f(1 + a + b) = 0

f(f(2)) = 0 $\Rightarrow$ f(4 + 2a + b) = 0

$\therefore$ 1 + a + b and 4 + 2a + b are roots of f(x)

$\therefore$ 3a + 2b + 5 = –a

$\Rightarrow$ 4a + 2b + 5 = 0 ..... (1)

(a + b + 1)(2a + b + 4) = b

$\Rightarrow (a+b+1)\left(-\dfrac{5}{2}+4\right)=b \Rightarrow (a+b+1)\left(\dfrac{3}{2}\right)=b$

$\Rightarrow$ 3a + 3b + 3 = 2b

$\therefore$ f(0) = b

$\Rightarrow$ b = –3a – 3 ..... (2)

From equation (1), we get

4a + 2(–3a – 3) + 5 = 0

$\Rightarrow$ –2a – 1 = 0 $\Rightarrow a=-\dfrac{1}{2}$

$\therefore\ b=-3\left(-\dfrac{1}{2}\right)-3=-\dfrac{3}{2},\ f(0)=b=-\dfrac{3}{2}$

41. A

Sol. $y=\tan^{-1}\left(\dfrac{5x-x}{1+5x(x)}\right)+\tan^{-1}\left(\dfrac{x+\frac{2}{3}}{1-\frac{2x}{3}}\right)=\left(\tan^{-1}(5x)-\tan^{-1}(x)\right)+\left(\tan^{-1}x+\tan^{-1}\left(\dfrac{2}{3}\right)\right)$

$\dfrac{dy}{dx}=\dfrac{5}{1+25x^2}$; $\lambda = 25$; $k = 5$; $\dfrac{\lambda}{k}=5$

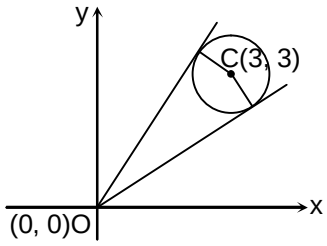
42. B

Sol. $y = mx$

$$\left|\frac{3m-3}{\sqrt{m^2+1}}\right| = \sqrt{6}$$

$\Rightarrow m^2 - 6m + 1 = 0$

$m_1 + m_2 = 6$



43. A, B

Sol. $a+b+c = -\frac{(\lambda+2)}{8}$; $ab+bc+ca = -\frac{(2k+\lambda)}{8}$; $abc = \frac{27}{8}$

$$(a+b+c)(ab+bc+ca) = \left(-\frac{(\lambda+2)}{8}\right)\left(-\frac{(2k+\lambda)}{8}\right) = \frac{2^3\cdot 3^5}{8\times 8} = \frac{3^5}{8}$$

$$\Rightarrow abc(a+b+c)\left(\frac{1}{a}+\frac{1}{b}+\frac{1}{c}\right) = \frac{3^5}{8} \Rightarrow \frac{27}{8}(a+b+c)\left(\frac{1}{a}+\frac{1}{b}+\frac{1}{c}\right) = \frac{3^5}{8}$$

$$\Rightarrow (a+b+c)\left(\frac{1}{a}+\frac{1}{b}+\frac{1}{c}\right) = 9$$

AM $\geq$ HM ; $\frac{a+b+c}{3} \geq \frac{3}{\left(\frac{1}{a}+\frac{1}{b}+\frac{1}{c}\right)} \Rightarrow (a+b+c)\left(\frac{1}{a}+\frac{1}{b}+\frac{1}{c}\right) \geq 9$

$a = b = c \Rightarrow a = \frac{3}{2}$; $2R = \frac{a}{\sin A} = \frac{3}{2\sin 60^\circ} = \frac{3}{2\times\frac{\sqrt{3}}{2}} = \sqrt{3}$; $R = \frac{\sqrt{3}}{2}$

$2R\cos B\cdot\cos C = 2R\times\frac{1}{2}\times\frac{1}{2} = 2\times\frac{\sqrt{3}}{2}\times\frac{1}{2}\times\frac{1}{2} = \frac{\sqrt{3}}{4}$

44. A, C

Sol. Let $3^k = x$; $2^k = y$

$$\frac{6^k}{(3^k-2^k)(3^{k+1}-2^{k+1})} = \frac{xy}{(x-y)(3x-2y)} = \frac{y(3x-2y)-2y(x-y)}{(x-y)(3x-2y)} = \frac{y}{x-y} - \frac{2y}{3x-2y}$$

$$= \frac{2^k}{3^k-2^k} - \frac{2^{k+1}}{3^{k+1}-2^{k+1}} \text{ ; } \sum_{k=1}^{n}\frac{6^k}{(3^k-2^k)(3^{k+1}-2^{k+1})}$$

$$= \frac{2}{3-2} - \frac{2^2}{3^2-2^2}; k=1$$

$$\frac{2^2}{3^2-2^2} - \frac{2^3}{3^3-2^3}; k=2$$

.....

$$\frac{2^n}{3^n-2^n} - \frac{2^{n+1}}{3^{n+1}-2^{n+1}}; k=n$$

$$\sum_{k=1}^{\infty}\frac{6^k}{(3^k-2^k)(3^{k+1}-2^{k+1})} = 2$$

45. A, B

Sol. (A) $AB = O$ where $\det(A) \neq 0 \Rightarrow A^{-1}$ exists
Now pre-multiplying with A^{-1}, we get $(A^{-1}A)B = A^{-1}O$
$\Rightarrow B = O$ null matrix

(B) $\det(3ABC) = 3^2 \det(A) \det(B) \det(C) = 9(2)(3)(4) = 216$
($\because$ A, B, C are square matrices of order 2)

(C) $\det(A) = \frac{1}{2}$ (order of matrix A is 3)

$\because \det(\text{adj } A) = (\det(A))^{n-1}$ place A by A^{-1} $\det(\text{adj } A^{-1}) = \left|A^{-1}\right|^2 = \frac{1}{|A|^2} = \frac{1}{\left(\frac{1}{2}\right)^2} = 4$

(D) Skew symmetric matrix of odd order is singular. But, if order of skew symmetric matrix is even, then it need not be singular.
For example, $A = \begin{bmatrix} 0 & -4 \\ 4 & 0 \end{bmatrix}$ and $\det(A) = 16$

46. A, C, D

Sol. ${}^{100}C_{50} = \frac{(100)!}{50!\,50!}$; $E_2(100!) = 97$; $E_2(50!) = 47$

$\therefore$ Exponent of 2 in ${}^{100}C_{50} = 3$

Similarly exponents of 3, 5, 7 in ${}^{100}C_{50}$ are 4, 0 and 0 respectively

$\therefore {}^{100}C_{50} = 2^3 \cdot 3^4 \cdot 5^0 \cdot 7^0$ ..... x = 3, y = 4, z = k = 0

47. A, B, C, D

Sol. $\left({}^{20}C_1 + {}^{20}C_3 + {}^{20}C_5 + {}^{20}C_7 + {}^{20}C_9\right)^5 = \left(2^{18}\right)^5 = 2^{90} = \left(2^5\right)^{18} = (33-1)^{18} \Rightarrow k = 1$

48. B, C, D

Sol. $z^k = \cos\left(\frac{k}{1000}\right) + i\sin\left(\frac{k}{1000}\right)$; $\sin\left(\frac{k}{1000}\right) > \frac{1}{2} \Rightarrow \frac{k}{1000} > \frac{\pi}{6}$

$\Rightarrow k > \frac{1000 \times 3.1416}{6} \Rightarrow k > 523$

SECTION – C

49. 00100.00

Sol. $f(0) = \pm 10$; $\underbrace{(f(x)+1)}_{\text{Rejected}}(f'(x)+1) = 0$; f(x) = 10 – x or –10 – x

$\therefore \frac{1}{|k|}\sum_{f(x)\in k} |f(100)| = \frac{1}{2}|90 + 110| = 100$

50. 00000.90

Sol. $\frac{9}{10} = 0.9$; $\vec{a}\cdot\vec{b} = \vec{b}\cdot\vec{c} = \vec{c}\cdot\vec{a} = \cos\theta$; $v = \frac{1}{6}\left[\vec{a}\ \vec{b}\ \vec{c}\right]$

$\left[\vec{a}\ \vec{b}\ \vec{c}\right]^2 = \begin{vmatrix} 1 & \cos\theta & \cos\theta \\ \cos\theta & 1 & \cos\theta \\ \cos\theta & \cos\theta & 1 \end{vmatrix}$; $36v^2 = (1 + 2\cos\theta)(1 - \cos\theta)^2$

$$36\times\frac{1}{360}=(1+2\cos\theta)(1-\cos\theta)^2 \Rightarrow 3\cos^2\theta-2\cos^3\theta=\frac{9}{10}$$

51. 00016.00

Sol. $x\frac{d}{dx}\left(\ln\left(f\left(g(x)\right)\right)\right)=\frac{d}{dx}\left(\ln\left(g(f(x))\right)\right)$

$\int_0^p f\left(g(x)\right)dx=\frac{1}{2}-\frac{e^{-2p}}{2} \Rightarrow f(g(p))=e^{-2p}$

$\therefore\ x\frac{d}{dx}\ln\left(e^{-2x}\right)=\frac{d}{dx}\ln\left(g(f(x))\right) \Rightarrow -2x=\frac{d}{dx}\left(\ln\left(g(f(x))\right)\right)$

$\Rightarrow \ln(g(f(x)))=-x^2+c$

$g\left(f(x)\right)=k'\cdot e^{-x^2}$; $g(f(0))=1 \Rightarrow k'=1$; $g(f(4)))=e^{-16}$

$\Rightarrow k=16$

52. 00000.50

Sol. $\because$ Ellipse contains the circle

$\therefore b^2x^2+a^2(1-(x-1)^2)=a^2b^2 \Rightarrow (b^2-a^2)x^2+2a^2x-a^2b^2=0$

$\therefore D=0$; $4a^4+4(a^2b^2)(b^2-a^2)=0 \Rightarrow a^2+b^2(b^2-a^2)=0$

$\Rightarrow a^2-b^2\,a^2e^2=0 \Rightarrow a^2(1-b^2e^2)=0 \Rightarrow 1=b^2e^2 \Rightarrow be=1$

Area of ellipse $A=\pi ab$; $A^2=\pi^2a^2b^2$; $e^2=\frac{1-b^2}{a^2}=\frac{1}{b^2} \Rightarrow a^2=\frac{b^4}{b^2-1}$

$A^2=f(b)=\frac{b^6}{b^2-1}$; $f'(b)=0 \Rightarrow (b^2-1)(6b^5)-b^6\times 2b=0$

$\Rightarrow 3\left(b^2-1\right)=b^2 \Rightarrow b^2=\frac{3}{2}$; $a^2=\frac{9}{2}$

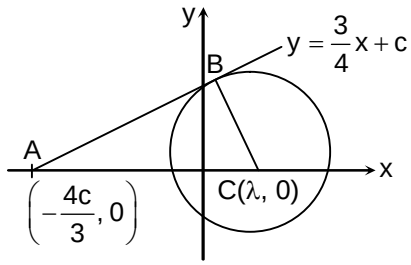
$a^2+b^2=\frac{9}{2}+\frac{3}{2}=\frac{12}{2}=6$; $12k=6 \Rightarrow k=\frac{1}{2}=0.5$

53. 00023.00

Sol. $|AB|=\sqrt{\left(\lambda+\frac{4c}{3}\right)^2-(5)^2}$

$=\sqrt{\left(\frac{3\lambda+4c}{3}\right)^2-25}$

$=\sqrt{\left(\frac{25}{3}\right)^2-25}=\frac{20}{3}$



54. 00000.80

Sol. Equation of line through O(0, 0, 0) and perpendicular to the plane $2x-y-z=4$, is

$\frac{x-0}{2}=\frac{y-0}{-1}=\frac{z-0}{-1}=\lambda\ (\text{say})$

$\therefore$ Any point on line is $(2\lambda, -\lambda, -\lambda)$

$\therefore$ Above point lies on the plane $3x-5y+2z=6$, so $6\lambda+5\lambda-2\lambda=6 \Rightarrow \lambda=\frac{2}{3}$

$\therefore$ Coordinates of point of intersection is $\left(\frac{4}{3},-\frac{2}{3},-\frac{2}{3}\right)=(\alpha,\beta,\gamma)$

$\therefore 2\alpha-3\beta+\gamma=4$